GEOGRAPHY

INDIAN GEOGRAPHY





INDIAN SOILS

MODULE - 8

PROBLEMS OF INDIAN SOILs

PROBLEMS OF INDIAN SOILS

2 SOIL SALINITY AND SOIL ALKALINITY

Salinity creeping into croplands

1. Soil Salinity and Soil Alkalinity are the results of over irrigation in small irrigated areas.

2. In canal irrigated areas, plenty of water is available and the farmers indulge in over irrigation of their fields.

3. Under such conditions, the ground water level rises and saline and alkaline efflorescences consisting of salts of sodium, calcium and manganese appear on the surface as a layer of white salt through capillary action.

PROBLEMS OF INDIAN SOILS

2 SOIL SALINITY AND SOIL ALKALINITY

4 Salinity means the predominance of chlorides and sulphates of sodium, calcium and magnesium in the soils in sufficient quantity to be able to seriously interfere with the growth of most plants.

5 Alkalinity implies the dominance of sodium salts, specially sodium carbonate.

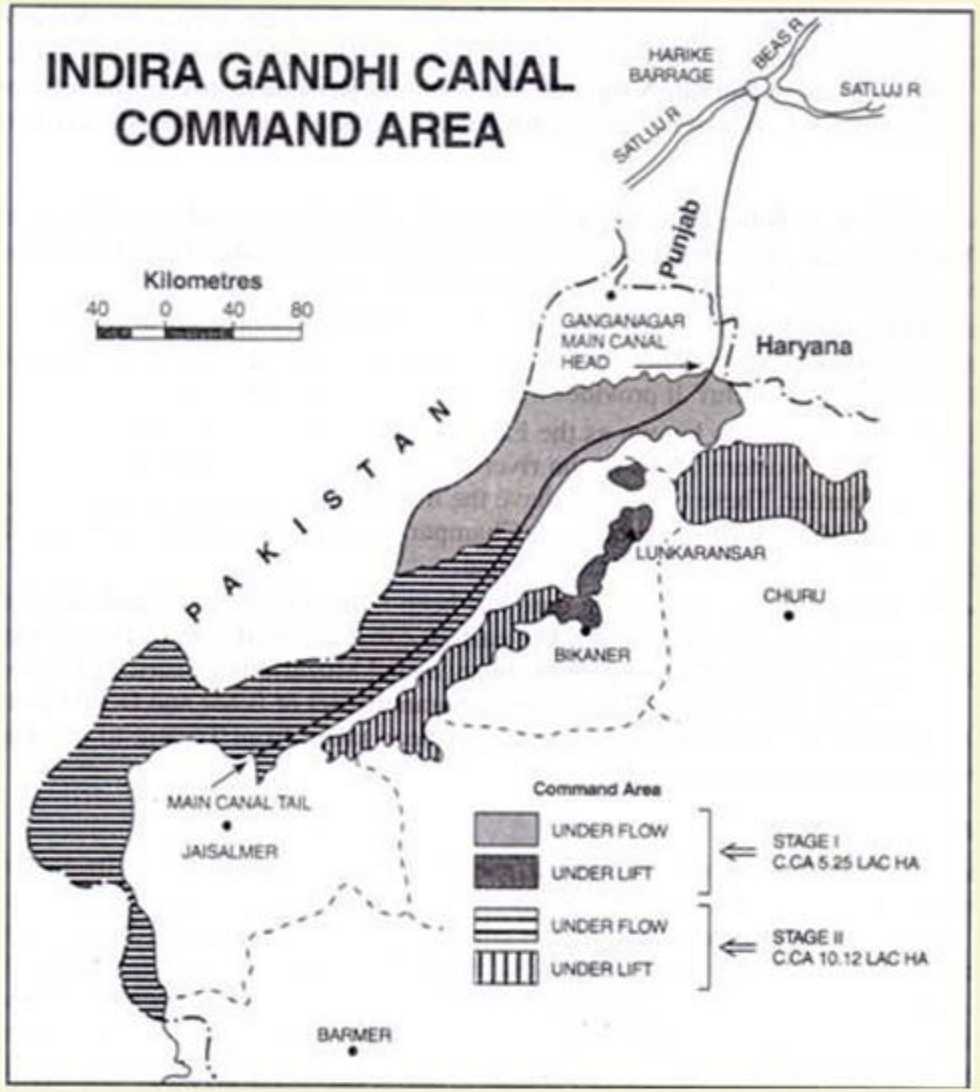
6 Increasing salinity and alkalinity always indicate extension of water logging salt encrustation (saline efflorescence) or thur tendencies.

7 Salinity and alkalinity have adverse effect on soil and reduce soil fertility

8 *Note: Saline soils are known by different names as reh, kallar, usar, thur, chopan etc.*

PROBLEMS OF INDIAN SOILS

2 SOIL SALINITY AND SOIL ALKALINITY



9 Vast tracts of canal irrigated areas in Uttar Pradesh, Punjab and Haryana; arid regions of Rajasthan, semi-arid areas of Maharashtra, Gujarat, Andhra Pradesh and Karnataka and coastal areas of Orissa, Gujarat, West Bengal are facing this problem.

10 Although Indira Gandhi Canal in Rajasthan has turned the sandy desert into granary, it has given birth to serious problem of salinity and to alkalinity.

11 In Gujarat, the area around Gulf of Khambhat is affected by sea tides carrying silt-laden deposits.

PROBLEMS OF INDIAN SOILS

2 SOIL SALINITY AND SOIL ALKALINITY

12 Soil salinity and alkalinity has many adverse effects, some important effects are as under:

a) Soil fertility is reduced which results in crop failure. Cultivation is not possible on saline soils unless they are flushed out with large quantities of irrigation water to leach out the salts.
b) Choice of crops is limited because some crops are sensitive to salinity and alkalinity. Only high salt tolerant crops such as cotton, rape, barley etc. and medium salt tolerant crops like wheat, rice, linseed, pulses, millets etc can be grown
c) Quality of fodder becomes poor.
d) Salinity and alkalinity create difficulties in building and road construction.
e) It causes floods due to reduced infiltration, leading to crop damage in the adjoining areas.

PROBLEMS OF INDIAN SOILS

2 SOIL SALINITY AND SOIL ALKALINITY

MEASURES TO TREAT SALINITY AND ALKALINITY

Following steps are necessary to treat salinity and alkalinity and restore the fertility of the soil:

1. Providing outlets for water to drain out excess water.

2. Minimizing the use of water. Making judicious use of irrigation facilities.

3. Planting salt tolerant vegetation and crops such as cotton, barley, date palm, linseed & certain grasses as fodder can be helpful.

4. It has been found that crop rotation involving Dhaincha (green fodder) – cotton, in the Deccan Plateau, Dhaincha – rice in the Uttar Pradesh and Dhaincha –rice- barseem in Punjab and Haryana have been very helpful.

PROBLEMS OF INDIAN SOILS

2 SOIL SALINITY AND SOIL ALKALINITY

MEASURES TO TREAT SALINITY AND ALKALINITY

Gypsum being applied on the fields

5. Liberal application of gypsum in upper 15cm thick soil to convert the alkalis into soluble compounds. Alkali can be removed by adding sulphuric acid or acid forming substances like sulphur and pyrite. Also organic residues such as rice husks and rice straw can be added to promote formation of mild acid as a result of their decomposition.

6. Flushing the salt by flooding the fields with excess water. However this practice can lead to accumulation of saline water in the downstream area.

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

1. Desertification can be defined as spread of desert like conditions in arid or semi-arid areas due to man's influence or climate change.
2. A large part of the arid and semi-arid region belonging to Rajasthan and south Punjab and Haryana, lying between the Indus and Aravali range is affected by desert conditions.
3. This area is characterised by scanty rainfall, sparse vegetation cover and acute scarcity of water.

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

Rajasthan desert

4 The Rajasthan desert is partly natural but largely man made and the fear is often expressed, quite rightly too, that the desert is spreading and that conditions within the desert are deteriorating.

5 Desert soils suffers maximum erosion by wind. The sand carried by wind is deposited on the adjoining fertile lands whose fertility dwindles and sometimes the fertile land merges with the advancing desert.

6 It has been estimated that the Thar Desert is advancing at an alarming rate of about 0.5 km per year.

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

Ladakh desert

7 Currently about 12.13 per cent of India's total area is classified as arid and about 30 % area is termed as semi arid.

8 The highest concentration of arid land is found in Rajasthan where more than half of the total area is arid.

9 Gujarat has about one-third of its total area is arid lane.

10 Jammu and Kashmir has cold deserts in Leh and Ladakh where the annual rainfall hardly exceeds 20 cm.

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

Rain shawdow effect of Western Ghats

11 Arid and semi-arid conditions prevail in large parts of Rajasthan, Gujarat, Punjab, Haryana, Maharashtra, Andhra Pradesh, Karnataka, Jammu and Kashmir, Uttar Pradesh, Tamil Nadu and the Madhya Pradesh.

12 Semi-arid conditions in north-western parts of India and in the States of Rajasthan, Punjab, Haryana etc are created by their long distances from the sea while such conditions in the southern States like Karnataka, Andhra Pradesh, Tamil Nadu, Maharashtra etc. are caused by the presence of the Western Ghats. These States are location in the rain shadow area of the Western Ghats.

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

13 The process of desertification is attributed to various causes of which more important are uncontrolled grazing, reckless felling of trees and growing population. Climate changes have also contributed to the spread of deserts.

Over-grazing

Deforestation

Climate change

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

14 Desertifications has several ecological implications. Some important implications are listed below.

i. Drifting of sand and its accumulation on fertile agricultural land.
ii. Excessive soil erosion by wind and to some extent by water.
iii. Deposition of sand in rivers, lakes and other water bodies thereby decreasing their water containing capacity.
iv. Lowering of water table leading to acute water shortage.
v. Increase in area under wastelands.
vi. Decrease in agricultural production.
vii. Increase in frequency and intensity of droughts.

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

MEASURES OF CONTROLLING DESERTIFICATION

A worker laying down hay in a grid pattern to stabilize a sand dune

1. Intensive tree plantation programme should be initiated.
2. Shifting sand dunes in Bikaner, Barmer, Churu, Jaisalmer and Jhunjhunum districts of Rajasthan cover an approximate area of 74 thousand sq km.
3. Central Arid Zone Research Institute, Jodhpur has suggested mulching them with different plant species. Mulches are put in small squares and serve as an effective physical barrier to the moving sand.

PROBLEMS OF INDIAN SOILS

3 DESERTIFICATION

MEASURES OF CONTROLLING DESERTIFICATION

Overgrazing is a major issue

4. Grazing should be controlled and new pastures should be developed.
5. Indiscriminate felling of trees should be banned.
6. Alternative sources of fuel can reduce the demand for fuel wood and save the trees from destruction hence checking the onward march of the desert.
7. Sandy and wastelands should be put to proper use by judicious planning.

PROBLEMS OF INDIAN SOILS

4 WATERLOGGING

1. The flat or nearly level surfaces and saucer-like depressions make the movement of surface water sluggish leading to accumulation of rain water which in turn results in warterlogging and flooding of the soil.

2. Waterlogged soils are soaked or saturated with water. Moreover, seepage from unlined channels or canal systems also leads to waterlogging in the contiguous arable lands.

3. It has been estimated that extent of waterlogged soils is about 12 million hectares in India: half of which lies along its coast and the other half in the inland area under existing or newly created irrigation command areas.

PROBLEMS OF INDIAN SOILS

4 WATERLOGGING

Salinity associated with waterlogging

4 Waterlogging is believed to be one of the chief causes of salinity. The problem of land reclamation under waterlogging conditions is a complex one and must therefore, be tackled with great care as the scheme involves huge expenditure.

5 Proper layout of drainage schemes is the only way to overcome the menace of waterlogging. The basic methods of removing excess water from waterlogged soils are

a) Surface Drainage and
b) Vertical Drainage

As seen in Module 7

MAJOR SOIL TYPES OF INDIA

8 SALINE & ALKALINE SOILS

CAPILLARY ACTION

- *Capillary action is the ability of a liquid to flow in narrow spaces without the assistance of, and in opposition to, external forces like gravity.*
- *The force behind capillary action is surface tension.*

4. In regions with low water table, the salts percolate into sub soil and in regions with good drainage, the salts are wasted away by flowing water.

5. But in places where the drainage system is poor, the water with high salt concentration becomes stagnant and deposits all the salts in the top soil once the water evaporates.

6. In regions with high sub-soil water table, injurious salts are transferred from below by the capillary action as a result of evaporation in dry season.

PROBLEMS OF INDIAN SOILS

4 WATERLOGGING

MEASURES OF CONTROLLING WATERLOGGING

a SURFACE DRAINAGE

Surface Drainage involves the disposal of excess water over ground surface through an open drainage system with an adequate outlet. Surface drainage is helpful where

i. Soils are deep with low infiltration rates,
ii. Intensity of rainfall is high
iii. Terrain is level to nearly level and
iv. The water table is high

PROBLEMS OF INDIAN SOILS

4 WATERLOGGING

MEASURES OF CONTROLLING WATERLOGGING

b VERTICAL DRAINAGE

- Any bore or well from which the underlying water is extracted is defined as vertical drainage.
- The success of vertical drainage depends upon the presence of favorable aquifer and water table for lifting the ground water on sustainable basis and the favorable quantity of water that could be re-utilized for irrigation purposes. Such conditions prevail in the Indo-Gangetic plain.
- In the Punjab-Haryana plain the water-table has been successfully lowered through tubewell pumping. As a consequence waterlogged saline lands could be reclaimed.

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